

The Kidney in Schistosomiasis

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INTRODUCTION

Schistosomiasis is a parasitic disease usually acquired by teenagers, often leading to complications that may extend into the fourth and fifth decades of life. It was known to the ancient Egyptians as “the bloody urine disease”¹ and is also known as *bilharziasis* in honor of its discoverer, Theodor Bilharz, the German physician who practiced in Egypt in the 1850s.

The life cycle of the parasite is shown in Fig. 56.1. Infection is acquired through contact with contaminated waters in ponds and slow-flowing canals. Cercariae enter through the skin or mucous membranes and migrate through the lymphatics and blood circulation into the portal or perivesical venous system, where they mature into sexually differentiated adult worms and live in almost continuous copulation. Females leave the males only to lay eggs, and, traveling against the blood flow, reach the rectal or bladder mucosa. The ova are driven out by visceral contraction during defecation or urination. Contact with fresh water within a couple of days allows the eggs to hatch, releasing miracidia, which infect specific snails. In this intermediate host, the organisms mature asexually into cercariae, which are eventually released to search for their definitive host, which is usually humans and occasionally apes and cattle. The snail demography defines the endemicity and frequency of schistosomiasis in different geographic regions and is influenced by temperature and humidity.²

About 230 million inhabitants (about 50% children) of 78 countries on five continents are infected, and an additional 700 million are at risk. Of the infected persons, 60% are symptomatic, 10% have serious sequelae, and 1% die of the disease each year, mainly in China, the Philippines, Brazil, northern Senegal, and Uganda. There has been a significant decline in prevalence during the second half of the past century, as a result of mass treatment with tartar emetic during the 1960s, the use of praziquantel during the 1980s, and the World Health Organization (WHO)-sponsored global mass praziquantel treatment program starting in the late 1990s. This has resulted in eradication of schistosomiasis from Japan and the Lesser Antilles islands; transmission was halted in Tunisia and significantly reduced in China, Egypt, Morocco, Saudi Arabia, Brazil, Venezuela, and Puerto Rico. However, certain schistosomal strains—up to 82% in Senegal—are resistant to praziquantel. Such strains are responsible for incremental prevalence in several countries and even spread to adjacent geographic regions in East and West Africa.

Of seven species that affect humans, three are responsible for almost all major morbidity from the disease: *Schistosoma haematobium* throughout Africa and adjacent regions; *Schistosoma mansoni* in Africa, South America, and the Caribbean; and *Schistosoma japonicum* in the Far East (Fig. 56.2).

S. haematobium affects the genitourinary (GU) tract, whereas *S. mansoni* and *S. japonicum* typically affect the colon and rectum and

spread upstream through the portal venous system to the liver, leading to periportal fibrosis. All three species may cause “metastatic” lesions when ova are driven by the bloodstream to the lungs, brain, spinal cord, heart muscle, eyes, and other sites.³ Overall morbidity is variable and depends on the virulence of the infective strains, host resistance, environmental factors, and standards of primary medical care. For example, chronic lower urinary tract disease among infected subjects was reported a few decades back to vary from 2% in Nigeria in the west of Africa to 52% in Tanzania in the east.⁴ An even wider range (0.6%–67%) was reported among children in sub-Saharan Africa.⁵

The kidneys are secondary targets of both *S. haematobium* and *S. mansoni* infection. New cases of *S. mansoni*-associated nephropathy are reported mainly in South America^{6,7} and of *S. haematobium*-associated urogenital disease in Africa.⁸

PATHOGENESIS

Schistosomes cause morbidity through two major mechanisms: (1) local reactions around the ova deposited in different tissues and (2) systemic effects attributed to the host’s response to circulating antigens released from the worms or the ova (Fig. 56.3).^{4,9}

The local reaction is a granulomatous cell-mediated immune response to soluble egg antigens diffusing out of trapped ova through micropores in the eggshell. The initial response is innate, being driven by tissue macrophages, and involves natural killer cells, neutrophils, and complement. This is followed by a specific immune response orchestrated by T helper (Th) lymphocytes. The schistosomal granuloma is composed of mononuclear cells, eosinophils, neutrophils, basophils, and fibroblasts, which are recruited and activated by a variety of Th lymphokines and by specific chemoattractants of parasitic origin (Fig. 56.4). These cells are involved in the elimination of the parasite by direct phagocytosis (monocytes), lymphocytotoxicity (T lymphocytes), antibody-dependent cytotoxicity (eosinophils), and antibody- and complement-dependent cytotoxicity (neutrophils). Later, the granuloma is modulated by gradual switching from Th1 to Th2 activation, largely mediated by a change in the monokine profile that favors release of modulatory cytokines, which are associated with a phenotypic change of the committed tissue macrophages. At this stage, the intensity of the inflammatory reaction is reduced, and progressive fibrosis is induced largely through the release of interleukin (IL)-5, IL-10, IL-13, somatostatin, and transforming growth factor (TGF)- β . Data from murine models incriminate additional T-cell subsets in promoting fibrosis, including Th9, Th17, and T-follicular helper cells (Tfh).¹⁰ Further on, tolerance to the parasite is achieved by an established population of regulatory T (T-reg) cells, which develop under the combined influence of host- and ova-derived mediators. With the final extinction of the inflammatory reaction, granulomas in

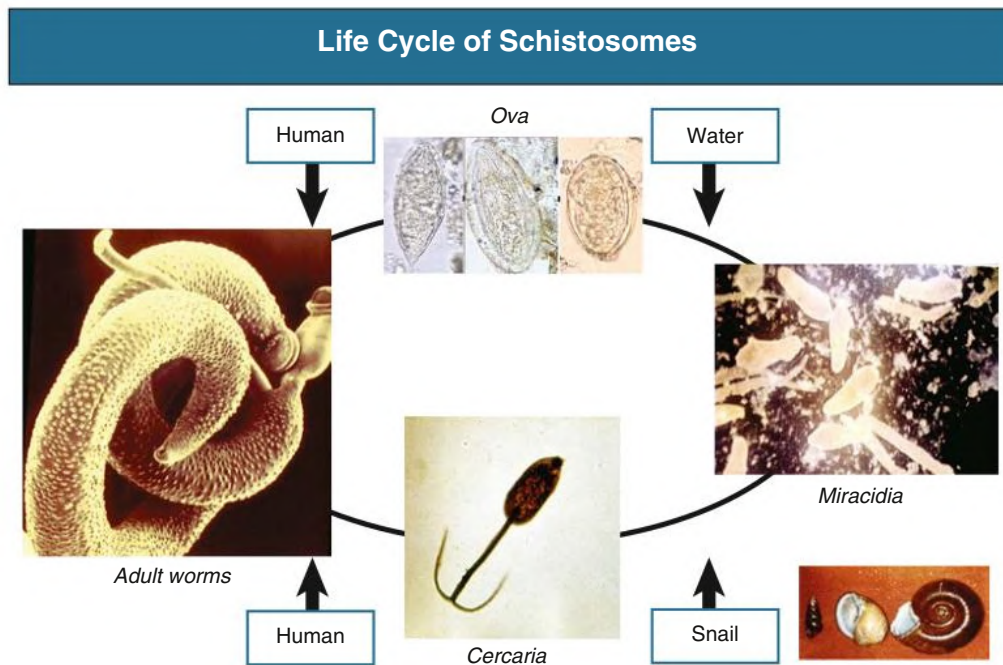


Fig. 56.1 Life cycle of schistosomes.

Global Distribution of Schistosomiasis

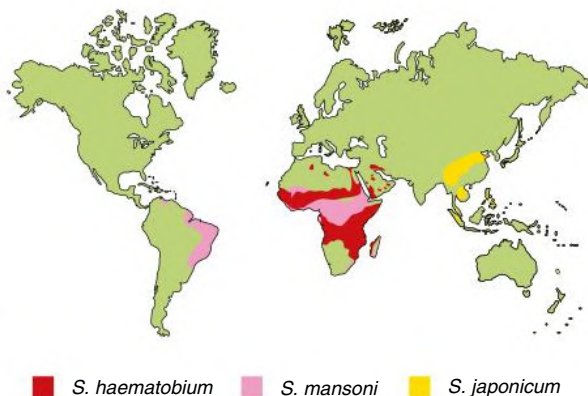


Fig. 56.2 World map showing the geographic distribution of the main pathogenic schistosomes.

the bladder, lower ureters, and seminal vessels heal with fibrosis and dystrophic calcification.^{11,12}

The systemic immune response is primarily a humoral reaction to circulating schistosomal antigens, which originate mainly from the worm's digestive enzymes (gut antigens), with a less significant contribution of tegument and oval proteins. The gut antigens consist of a positively charged glycoprotein and a negatively charged proteoglycan (circulating cationic antigens and circulating anionic antigens, respectively). These antigens are present in most of the schistosomal immune complex-mediated lesions, particularly in glomeruli.¹³ The antibody response is biphasic, driven by the successive Th1 and Th2 stages of lymphocyte activation. In the Th1 stage, B cells tend to produce immunoglobulin M (IgM), IgG1, and IgG3 under the influence of IL-2. During the Th2 phase, IgG2, IgG4, and IgA predominate; these have a limited ability to fix complement and may even block its deposition, hence their importance in modulating the granulomas.¹⁴

CLINICAL MANIFESTATIONS

Schistosomal lesions in the urinary system mirror the two major pathogenetic mechanisms. On one hand, there are local lesions, mostly affecting the lower urinary tract, caused by the local granulomatous response to *S. haematobium* ova.¹⁵ One may also observe lesions caused by immune complex deposition in the glomeruli, usually associated with hepato-intestinal *S. mansoni* and, less commonly, urogenital *S. haematobium* infections.¹⁵ Although glomerular lesions can be readily induced by *S. japonicum* in experimental animals,¹⁶ this species does not seem to cause significant kidney disease in humans.

Lower Urinary Tract Schistosomiasis

The lower urinary tract and adjacent genital structures are the primary sites of *S. haematobium* infection. Clinical disease starts by the coalescence of multiple granulomas that form small pseudotubercles in the bladder mucosa (Fig. 56.5). These may consolidate to form sessile, occasionally pedunculated masses or ulcerate, leading to painful terminal hematuria, the most typical presenting symptom. Hematuria may vary from microscopic (40%–100% in different reports) to gross (0%–97%).¹⁷ Ulcers eventually heal by fibrosis, with calcified granulomas under the atrophic and dirty mucosa, leading to the characteristic cystoscopic appearance of sandy patches and also the radiologic appearance of linear bladder calcifications in 2% to 62% of cases.¹⁷ Similar lesions may occur in the lower ureters, bladder neck, seminal vesicles, and other organs in the vicinity (Fig. 56.6).

The bladder lesions predispose to secondary bacterial infection, particularly with *Pseudomonas* or *Proteus* spp., usually after diagnostic or therapeutic instrumentation. *Proteus* infection is notorious for favoring stone formation, which further complicates the scenario. The subsequent fibrotic process may involve the bladder neck, leading to outflow obstruction, or the vesicoureteral junction, leading to ureteral obstruction or vesicoureteral reflux. Involvement of the detrusor is late and may lead to an atonic or a hyperirritable bladder. Eventually, the bladder becomes a deformed, contracted, and calcified organ that accommodates a very small amount of urine that is difficult to void.

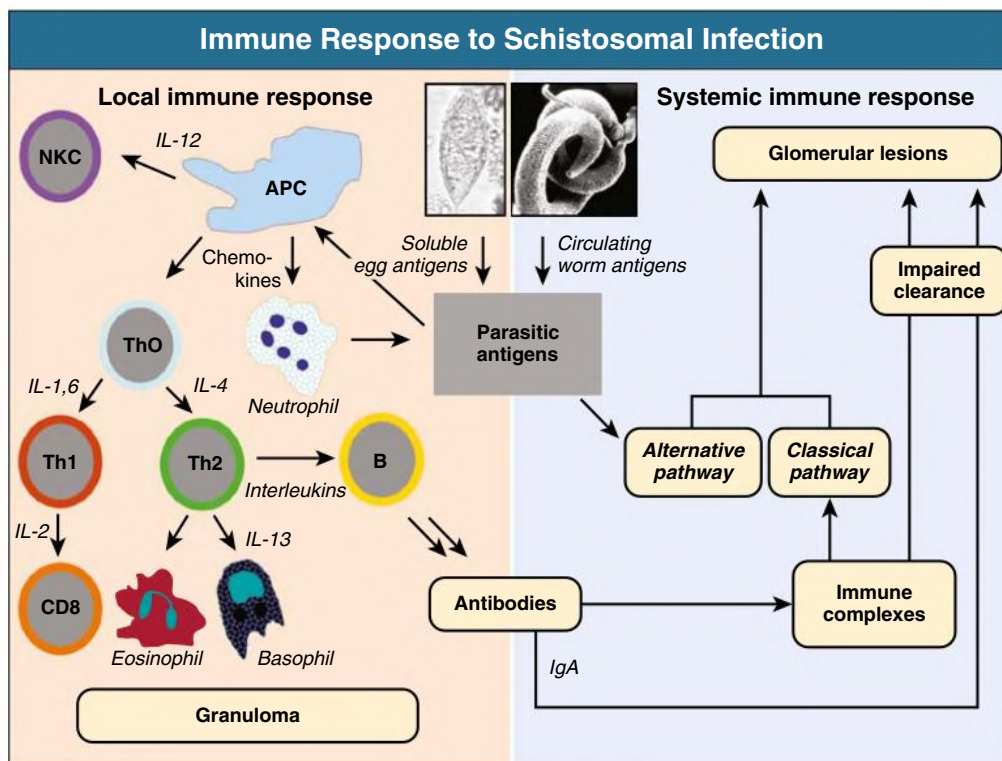


Fig. 56.3 Immune Response to Schistosomal Infection. The local immune response to deposited ova leading to granuloma formation is shown on the left; all cells shown in the diagram, in conjunction with antibodies or complement, participate in eventual parasite elimination (see text). The systemic immune response is shown on the right; note the important role of impaired clearance of schistosomal antigens and immunoglobulin A (IgA) in the development of glomerular lesions. APC, Antigen-presenting cell; IL, interleukin; NKC, natural killer cell.

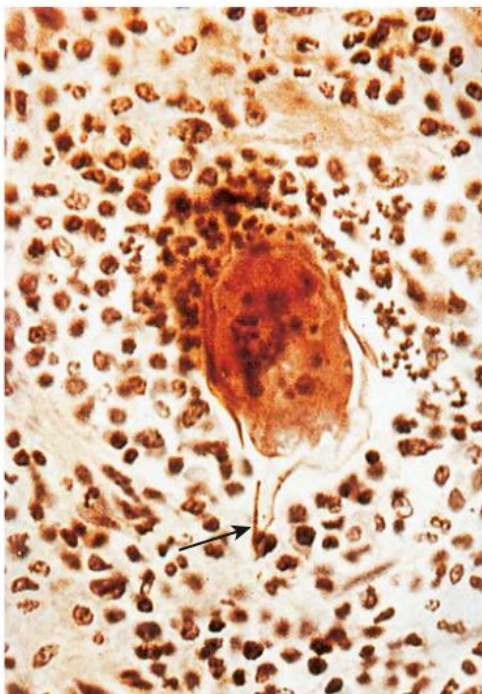


Fig. 56.4 *Schistosoma haematobium* Granuloma. Note the egg's terminal spike (arrow), which identifies *S. haematobium*, and the distortion of shell by proteases and oxidants released by the local neutrophil infiltration. (Hematoxylin-eosin stain, $\times 500$.)

Bladder Cancer

Chronic bilharzial cystitis is precancerous. The typical tumor is a squamous cell carcinoma,¹⁸ although a time trend to predominance of transitional cell carcinoma was reported in a large series of close to 10,000 Egyptian patients with bladder cancer.¹⁹ The tumor, particularly when of the squamous cell type, remains localized for a long time before spreading to the surrounding pelvic tissues or a distant site, thanks to the occlusion of lymphatics by the preceding fibrotic process.

Products of *S. haematobium* ova are directly incriminated in oncogenesis.^{20,21} The major impact is on the *BCL2* gene, which leads to epithelial proliferation and inhibition of apoptosis. This is coupled by inhibition of the tumor suppressor genes *P27*, *TP53*, and others. Other oncogenic factors in schistosoma-associated bladder cancer include disturbance of the urinary microbiome and secondary bacterial infection.²¹

Development of malignancy is suspected when the symptoms of chronic cystitis worsen, along with recurrence of hematuria many years after the initial presentation and the passage of small pieces of necrotic tissue with urine (necroturia). A characteristic radiologic sign is the irregular "eating up" of the bladder calcification on a plain radiograph (Fig. 56.7). Cystoscopy shows the tumor and provides the means for a histologic diagnosis.

Upstream Consequences

Although ureteral strictures and calcifications are common, the hypertrophied upper ureteral musculature usually overcomes the lower obstruction, thereby limiting the upstream consequences. Nevertheless, hydronephrosis and progressive kidney failure may

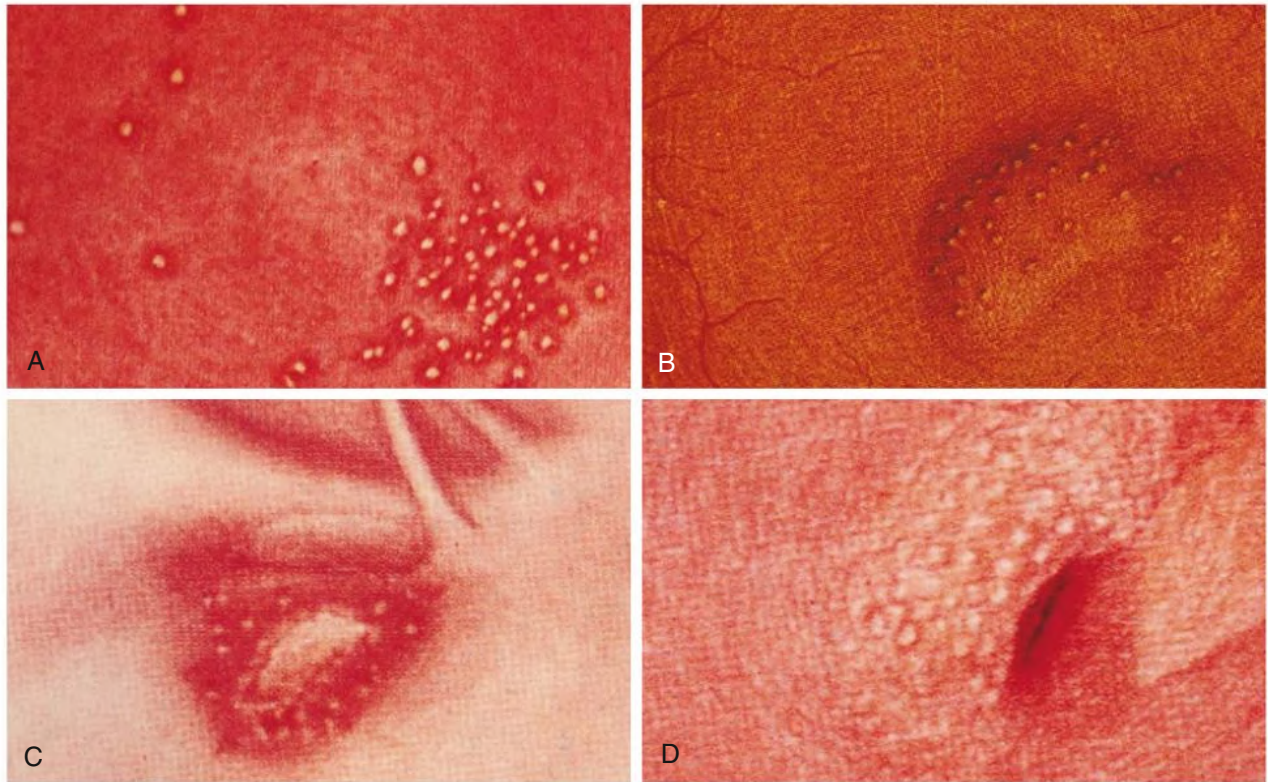


Fig. 56.5 Hand-Painted Cystoscopic Appearances in Urinary Schistosomiasis. (A) Pseudotubercles. (B) Sessile mass covered by pseudotubercles. (C) Ulcer surrounded by pseudotubercles. (D) Sandy patches. (Courtesy Professor Naguib Makar.)

develop when there is extensive ureteral scarring, in the presence of stones or secondary bacterial infection, or when the vesicoureteral junction is incompetent (Fig. 56.8). The frequency of upper urinary tract disease from *S. haematobium* is variably reported from different geographic regions—for example, from less than 10% in Niger to 48% in Cameroon.⁴

Interstitial Nephritis

Progressive kidney scarring is the eventual outcome of complicated *S. haematobium* infection, resulting from obstruction, reflux, and bacterial infection. Granulomas have occasionally been seen in the kidney interstitium but tend to be discrete and of no functional significance (Fig. 56.9).

The typical pathologic picture is that of a deformed kidney with calyceal dilation, distortion, and atrophic parenchyma. Dense interstitial infiltration, fibrosis, and periglomerular scarring are present. The glomeruli may show ischemic collapse or other schistosomal lesions, such as proliferative glomerulonephritis (GN) or amyloidosis.

The clinical picture is that of chronic tubulointerstitial nephritis (see Chapter 65), often associated with residual manifestations of lower urinary tract involvement. Hypertension is a late feature, being checked by concomitant tubular salt wasting. Anemia and osteodystrophy may be disproportionately severe because of the associated secondary distal tubular acidosis and nutritional deficiency in endemic areas.

Immune-mediated tubulointerstitial nephritis has been described with *S. mansoni* in humans.^{22,23} Tubular dysfunction associated with hepatosplenic schistosomiasis includes salt wasting, impaired concentrating ability,²⁴ and Type I renal tubular acidosis.^{24,25} Because these were also reported in chronic liver disease from other causes, their specificity to schistosomal etiology remains questionable.

Glomerulonephritis

Immune complex–mediated GN has been described in experimental¹⁶ and human¹⁵ infection, mainly with *S. japonicum* and *S. mansoni*. The latter species accounts for most clinically significant disease in humans. *S. haematobium* GN is rare, transient, and usually subclinical.²⁶

Schistosomal gut antigens are present in circulating immune complexes and in the immune deposits in mesangial, subendothelial, and intramembranous locations. The presence of liver fibrosis is critical because it results in impaired hepatic clearance of schistosomal antigens and immune complexes. These are mostly generated within the portal venous system, which accommodates the adult worms.

Most patients are 20- to 40-year-old men with evidence of hepatosplenic schistosomiasis. This is often discovered accidentally during a physical or ultrasonographic abdominal examination in an asymptomatic case. Less frequently, these patients may complain of abdominal distension, edema, upper gastrointestinal (GI) bleeding, or manifestations of chronic liver disease. Kidney involvement is asymptomatic in up to 40% of those, being identified by accidental or surveillance urinalysis, which may display various grades of proteinuria, or abnormal sediment. Some 15% of those with hepatosplenic schistosomiasis have overt GN, with proteinuria and microhematuria on presentation and with or without the nephrotic syndrome, hypertension, and impaired kidney function. Liver function test results are often normal. A polyclonal gammopathy is seen in most cases, whereas a monoclonal IgM response is seen in those with associated hepatitis C virus (HCV) infection and cryoglobulinemia. Rheumatoid factor activity and anti-DNA antibodies are detected in 5% to 10% of cases, particularly in association with *Salmonella* infection (see later discussion), but they do not correlate with clinical severity. Rheumatoid factor seropositivity is detected in much higher titers when HCV infection is associated.²⁷

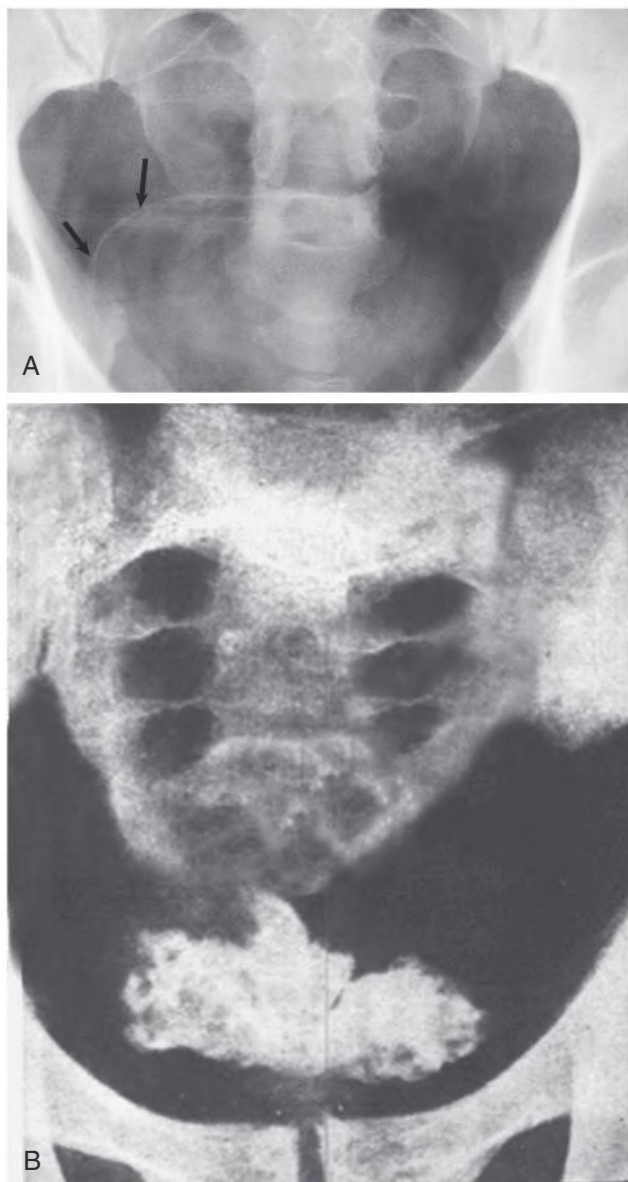


Fig. 56.6 Plain radiographic Appearances in *Schistosoma haematobium* Infection. (A) Faint linear calcification of the bladder wall (arrows). (B) Dense calcifications of contracted bladder and seminal vesicles.

Six histologic classes of schistosomal GN are recognized (Table 56.1 and Fig. 56.10).^{15,27} Class I (mesangial proliferative), class III (membranoproliferative), and class IV (focal segmental proliferative and sclerosing) result from the deposition of immune complexes representing different stages in the evolution of pure schistosomal hepatosplenic disease. The main deposits in class I are schistosomal antigens, IgM, and C3; and in class III and class IV, IgG and IgA, usually without schistosomal antigens. The IgA deposits parallel the severity of proteinuria and mesangial proliferation. Impaired hepatic clearance and increased mucosal synthesis of IgA have been documented in many of those patients.²⁸ Whereas class I lesions are seen in most asymptomatic cases, class III and class IV are usually symptomatic and progressive, even with eradication of the parasitic infection.

Class II (diffuse proliferative and exudative) is associated with urinary or biliary coinfection with *Salmonella* strains, usually *Salmonella paratyphi* A in Africa and *Salmonella typhimurium* in Brazil, which are attached to specific receptors in the tissues of adult schistosomes. C3 and *Salmonella* antigens have been detected in the glomerular capillary



Fig. 56.7 Schistosomal bladder cancer.



Fig. 56.8 Ascending cystogram showing right megaureter caused by vesicoureteral reflux.

walls and the mesangium. In these patients the clinical presentation is typical of acute postinfectious GN, associated with manifestations of *Salmonella*-related toxemia (fever, exanthema, and severe anemia).

Serum amyloid A (AA) protein deposits are detected by special stains or electron microscopy in up to 15% of biopsy samples from patients with class III and class IV, whereas AA-amyloidosis may be

the predominant lesion (class V) in less than 5% of patients with clinically overt schistosomal GN. It occurs with heavy, often mixed infection regardless of the anatomic site. Minimal amyloid deposits do not seem to alter the clinical presentation or prognosis, but typical class V lesions are grossly nephrotic and relentlessly progressive.

Class VI lesion was described in patients with hepatosplenic schistosomiasis and HCV infection.²⁷ The lesion consists of a mixture of mesangial proliferation, amyloid deposition, fibrinoid necrosis, and

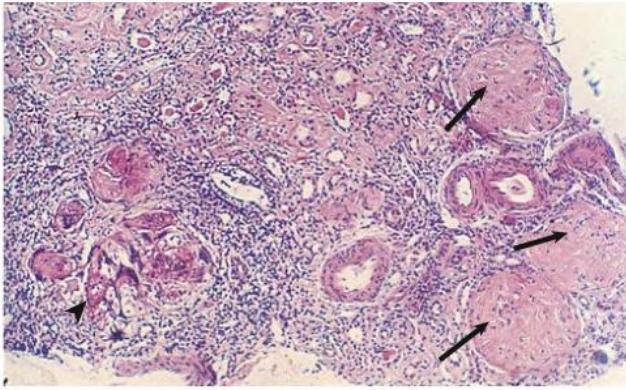


Fig. 56.9 Schistosomal Chronic Interstitial Nephritis. Note the dense cellular infiltration and fibrosis, atrophic dilated tubules, and thickened vessels. Three glomeruli heavily infiltrated with amyloid are seen on the right side (arrows); a schistosomal granuloma is seen in the lower left corner (arrowhead). (H&E stain; original magnification $\times 75$.)

cryoglobulinemic thrombi in the glomerular capillaries with tubular casts. Patients have chronic hepatitis, cirrhosis, nephrotic syndrome, cryoglobulinemic skin vasculitis, polyarthritis, and rapidly progressive kidney failure associated with severe protein-calorie malnutrition.

Other histologic lesions have been described in chronic schistosomiasis, yet a cause-and-effect relationship has not been established. A frequently debated lesion is membranous nephropathy, which is believed to be a chance association of primary membranous nephropathy and schistosomiasis.²⁹ Collapsing glomerulopathy has been described in a patient with APOL1 genotype infected with *Schistosoma mansoni*,³⁰ which was blamed as a triggering factor in a predisposed patient.

COINFECTION

Patients living in endemic areas may acquire one or more superimposed infections that can perturb schistosomal pathogenicity and modify its presentation. Mutually, schistosomiasis may modify the course of the confounding infection. The first reported coinfection is that of schistosomiasis and salmonellosis,³¹ discussed earlier as class II schistosomal glomerulopathy. Likewise, class VI is attributed to coinfection with HCV.²⁷

The list of other infections known to interact with schistosomiasis is expanding (Box 56.1). Some of the documented coinfections, namely those with plasmodium, filaria, tuberculosis, and staphylococci, are clinically significant. However, they were not reported to modify the known kidney lesions of schistosomiasis alone.

TABLE 56.1 Classification of Schistosomal Glomerulonephritis

Class	Histology	Immunofluorescence	Etiologic Agent	Prevalence	Clinical Findings	Treatment of Kidney Disease
I	Mesangial proliferative GN	IgM, C3, schistosomal antigens	<i>S. haematobium</i> , <i>S. mansoni</i> , <i>S. japonicum</i>	27%–60% of asymptomatic patients, 10%–40% of patients with kidney disease	Microhematuria, proteinuria	May respond to antiparasitic treatment
II	Diffuse proliferative exudative GN	C3, <i>Salmonella</i> antigens	<i>S. haematobium</i> , <i>S. mansoni</i> , plus <i>Salmonella</i> spp.	<i>Salmonella</i> infections	Acute nephritic syndrome, toxemia, reduced serum C3	May respond to treatment of <i>Salmonella</i> and schistosomal infections
III	Membranoproliferative GN	IgG, IgA, C3, schistosomal antigens	<i>S. mansoni</i> (<i>S. haematobium</i> ?)	7%–20% of asymptomatic patients and in 80% of patients with overt kidney disease	Hepatosplenomegaly, nephrotic syndrome, hypertension, kidney failure	No
IV	Focal segmental glomerulosclerosis	IgM, IgG (occasionally IgA)	<i>S. mansoni</i>	11%–38%	Hepatosplenomegaly, nephrotic syndrome, hypertension, kidney failure	No
V	Amyloid	AA protein	<i>S. mansoni</i> , <i>S. haematobium</i>	16%–39%	Hepatosplenomegaly, nephrotic syndrome, hypertension, kidney failure	No
VI	Cryoglobulinemic GN	IgM, C3	<i>S. mansoni</i> + hepatitis C virus	Unknown	Hepatosplenomegaly, nephrotic syndrome, purpura, vasculitis, arthritis, hypertension, kidney failure	Interferon plus ribavirin, corticosteroids, immunosuppression, plasma exchange

AA, Amyloid A; GN, glomerulonephritis; Ig, immunoglobulin.

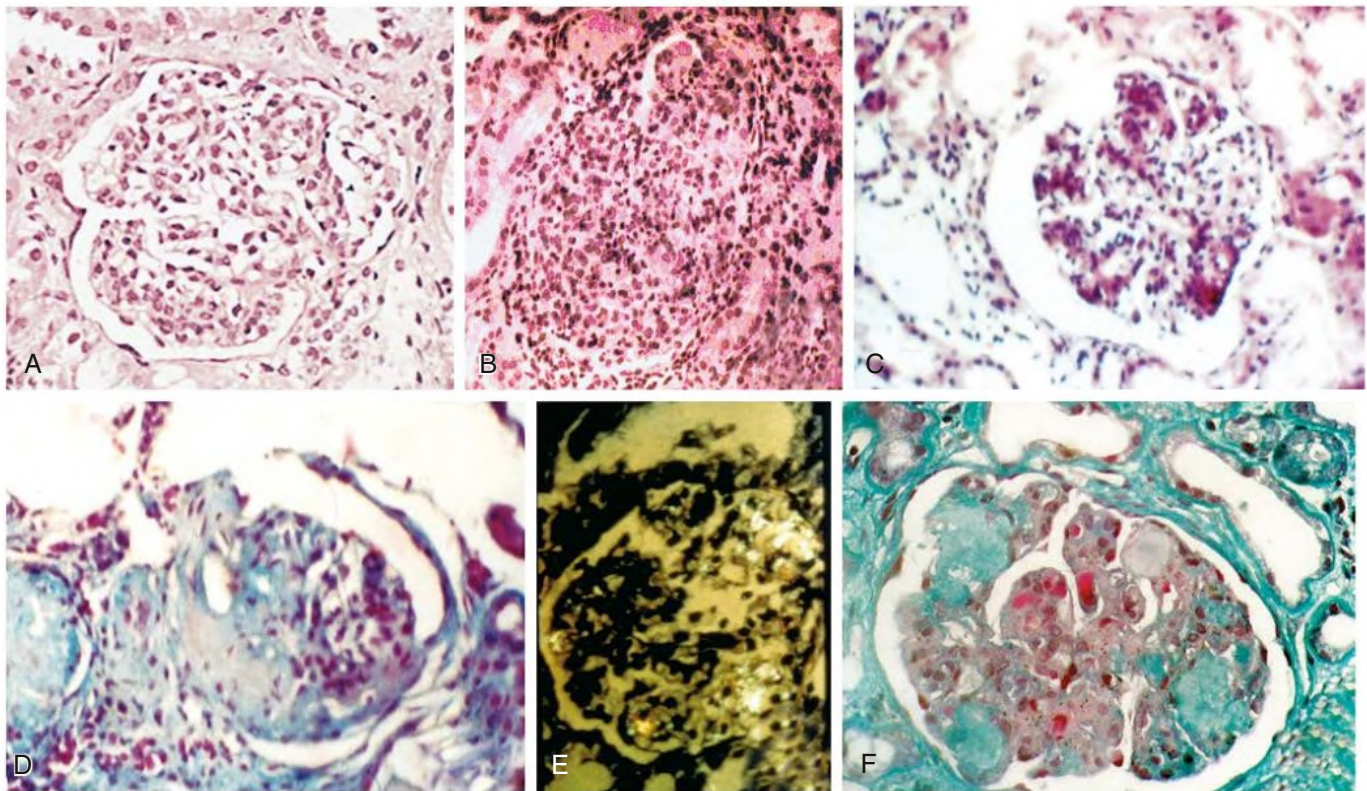


Fig. 56.10 Schistosomal Glomerulopathy. (A) Mesangial proliferative glomerulonephritis (GN), H&E stain (class I). (B) *Schistosoma*- and *Salmonella*-associated exudative GN, H&E stain (class II). (C) Membranoproliferative (mesangiocapillary) GN type I, H&E stain (class III). (D) Focal segmental glomerulosclerosis, Masson trichrome stain (class IV). (E) Green birefringence under polarized light in a glomerulus with mesangial proliferation in a patient with mixed *Schistosoma haematobium* and *Schistosoma mansoni* infection, Congo red stain (class V). (F) Amyloid deposits and cryoglobulin capillary thrombi (red stain) in a glomerulus displaying focal mesangial proliferation in a patient with schistosomal hepatic fibrosis and associated hepatitis C virus infection (class VI).

BOX 56.1 Clinically Relevant Coinfection with Schistosomiasis

Viral

Hepatitis B virus
Hepatitis C virus
Human immunodeficiency virus
Human papillomavirus

Bacterial

Tuberculosis
Staphylococci
Salmonella

Parasitic

Plasmodium
Filaria

On the other hand, there is evidence that coinfection with human immunodeficiency virus (HIV), hepatitis B virus (HBV), or human papilloma virus (HPV) may augment schistosomal morbidity:

Human Immunodeficiency Virus

Several studies have established an epidemiologic link between schistosomiasis and HIV infection. Females with lower urinary tract

schistosomiasis are at significantly increased risk (odds ratio [OR], 2.9 in Zimbabwe³² and 4.0 in Tanzania³³) of acquiring and subsequently transmitting HIV to their sexual partners. Data on males with *S. haematobium* infection or either sex with *S. mansoni* infection are inconsistent. There are no published data on *S. japonicum* and HIV coinfection.

The immunologic perturbation induced by schistosomiasis seems to accelerate the progression and spread of HIV infection.³⁴ HIV coinfection impairs the immune response, resulting in alteration of the granulomatous process involving schistosomes, yet without known clinical sequelae. However, HIV-infected patients are more vulnerable to reinfection with schistosomes upon new exposure after successful eradication.³⁵

There is some concern about the unexplained 3- to 16-fold increase of HIV viral load after treatment of schistosomiasis by praziquantel.³⁶ Fortunately, this effect is spontaneously reversible in a few weeks, and the benefit of eradicating the confounding parasitic infection on the clinical course and spread of HIV infection is unquestionable.³⁶

Hepatitis B Virus

Vulnerability of patients with hepatosplenic schistosomiasis to coinfection with HBV is well documented in hospital-based,³⁷ but not in community-based,³⁸ studies, perhaps because of the selection of complicated cases in the former. Such coinfection augments the liver injury induced by schistosomiasis,³⁹ leading to increased incidence of abnormal liver function (46.6%), jaundice (23.3%), hepatocellular carcinoma (17.8%), and liver-related mortality (23.3%).⁴⁰ It is unknown

whether schistosoma and HBV coinfection influences the nature or severity of the kidney lesions associated with either.

Human Papilloma Virus

The association of HPV infection with nonschistosomal squamous cell carcinoma of the bladder was reported in 1994.⁴¹ Three years later, an in-situ hybridization study in South Africa did not show evidence of such infection in 25 patients with bilharzial squamous cell bladder cancer.⁴² However, because of the confirmed association of HPV with cervical carcinoma in patients with genital schistosomiasis,⁴³ debate continues whether it may also be a confounding risk factor for bilharzial bladder cancer.

Diagnosis

Schistosoma haematobium Urinary Tract Disease

The clinical diagnosis of lower urinary tract schistosomiasis is easily made, particularly in patients with the typical pattern of painful terminal hematuria after exposure to fresh river waters in an endemic area. Diagnosis is more difficult when the history of exposure is less convincing (e.g., swimming pools) or when the clinical presentation is atypical (e.g., bacterial pyelonephritis, typhoid, or amyloidosis).

The diagnosis is made by finding ova in a fresh urine sample, which is easy because of their abundance, large size, and typical appearance (see Fig. 56.4). Live ova, containing mobile miracidia, indicate active infection, whereas dead, calcified ova may continue to be shed from fibrotic lesions for many months or even years.

Serologic diagnosis is based on finding circulating schistosomal antigens or antibodies.⁴⁴ The circumoval precipitin test is most frequently used in clinical laboratories. Serologic diagnosis is useful for diagnosis in the absence of ova, which occurs with old infections when the worms are sterile but continue to release their antigens. Serologic tests are also useful in assessing the response to treatment because titers usually become negative within 3 to 6 months of complete eradication of infection.

A polymerase chain reaction (PCR) real-time assay is available for detection of the *dra-1* *S. haematobium* antigen in the serum.⁴⁵ Because of its high cost, its clinical use is limited to early detection of infection in expatriates visiting endemic areas.

The radiologic appearances of bladder and seminal vesicle calcification are so typical that no further confirmatory tests are needed in an endemic area. Cystoscopic findings are equally pathognomonic although seldom required. Early pseudotubercles are easily distinguished from mycobacterial infection by their size and the surrounding mucosal pathologic changes. The presence of sandy patches with associated masses, polyps, and even neoplasms confirms the diagnosis. Tissue biopsy confirms the parasitic nature of the lesions. Different imaging techniques (e.g., ultrasound,⁴⁶ intravenous urography, voiding cystography) are useful in the diagnosis of upstream complications, including obstruction and reflux (see Chapters 61 and 62).

The main differential diagnosis for urinary schistosomiasis is tuberculosis, which also causes hematuria, strictures, back pressure, and chronic kidney disease (CKD). This differential can be resolved with appropriate parasitologic and bacteriologic techniques (see Chapter 54).

SCHISTOSOMA MANSONI GLOMERULONEPHRITIS

Overt glomerular disease in patients with hepatosplenic schistosomiasis is suspected in those who develop hypertension, nephrotic or nephritic syndrome, or CKD. Occult glomerular disease is indicated by the presence of abnormal urinary sediment or kidney function. Although kidney biopsy is essential for diagnosis and classification, no lesion is pathognomonic unless schistosomal antigens are detected,

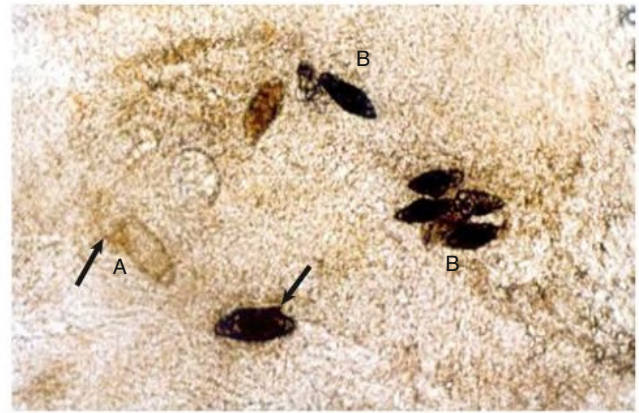


Fig. 56.11 Stool smears showing living (A) and dead (B) *Schistosoma mansoni* ova. Species are identified by the lateral spike (arrows).

which is unusual in clinically overt cases when sought with conventional immunofluorescence.

Identification of *S. mansoni* eggs in the stools (Fig. 56.11) or sub-mucosal rectal biopsy samples supports the diagnosis. Serologic tests for schistosomal gut antigens⁴⁷ and molecular tests for Sm1-7 encoding DNA⁴⁸ are usually positive. However, it is important to remember that these tests only indicate schistosomal infection and do not necessarily implicate schistosomes as causing the kidney disease.

Several biomarkers have been proposed for the detection of kidney injury in schistosomiasis. So far, none of these has reached a reliable, more specific, and noninvasive alternative to kidney biopsy.⁴⁹

Concomitant *Salmonella* or HCV infection can be detected by appropriate microbiologic tests. The various serologic abnormalities described are of limited diagnostic value, except the high rheumatoid factor, monoclonal IgM expansion, and low C4 that are typical of class VI.

Other glomerular disorders associated with hepatic fibrosis, such as secondary IgA nephropathy and hepatic glomerulosclerosis, should be considered in the differential diagnosis. However, in both these conditions, the kidney disease is relatively milder, mainly with microhematuria at presentation but rarely with nephrotic-range proteinuria or impaired kidney function. The glomerular deposits are mostly mesangial, in contrast to those seen in schistosomiasis, in which sub-endothelial and intramembranous deposits may also be present.

TREATMENT

Schistosoma haematobium Urinary Tract Disease

S. haematobium is susceptible to antimony compounds, organophosphates (metrifonate), and niridazole. However, the standard drug of choice is praziquantel, with a cure rate greater than 85% and the least toxicity. It is administered as a single oral dose of 40 mg/kg body weight, which may be repeated 2 weeks later if there is evidence of active disease. The drug has a low toxicity profile, mainly a mild to moderate transient elevation of hepatic transaminases. In infections resistant to praziquantel, the antimalarial *Artemisia annua* may be used as an alternative with equal effectiveness.⁵⁰ Antiparasitic treatment cures the early bladder disease but has no effect on sandy patches or other fibrotic lesions. Ureteral distension with radiologic evidence of hydronephrosis may be reversed in a few weeks after successful treatment.

Antibacterial therapy usually controls acute episodes of cystitis and pyelonephritis. However, it must be combined with simultaneous eradication of parasitic infection if still active, especially when the urinary bacterial infection is because of typhoid (*Salmonella typhi*).

Chronic fibrotic lesions are difficult to treat. Surgery or the placement of stents may be necessary for the relief of an obstructive lesion. However, particular caution is required in dealing with the vesicoureteral junction to avoid induction of reflux. Several plastic procedures are available to restore the distorted ureteral, bladder, or urethral anatomy. Associated bacterial infections may require long-term low-dose antibiotics.

Chronic dialysis in such patients can be difficult because of the negative effects of the associated schistosomal lesions in the liver, lungs, and other organs and the comorbid impact of undernutrition, viral infection, or malignant disease. The same factors can compromise outcomes of kidney transplantation, with the additional risk for urinary leakage, which is many-fold higher than usual because of the presence of fibrotic granulomas and anatomic distortion in the bladder wall. Reinfection with *S. haematobium* also has been described.⁵¹

Schistosoma Mansoni Glomerulonephritis

S. mansoni is more resistant to treatment and may require higher doses of praziquantel (40–60 mg/kg body weight) or the use of oxamniquine (single dose of 15 mg/kg body weight in South America or two doses of 15 mg/kg body weight given 12 hours apart in Africa). Research is underway to overcome oxamniquine resistance, which is limiting its use in many geographic regions.⁵² Nevertheless, eradication of the parasite can be curative only in classes I and II. In the latter, it must be combined with antibiotics for the control of *Salmonella* infection

(usually quinolones, macrolides, or third-generation cephalosporins). Antischistosomal and immunosuppressive therapy are ineffective in all other classes of schistosomal GN.¹⁵ These tend to progress to kidney failure in about 35% of cases 5 years after initial diagnosis.⁶

Chronic dialysis may be complicated by the risk for bleeding from esophageal or gastric varices after anticoagulation. Endoscopy is essential before the start of regular hemodialysis, with prophylactic sclerotherapy if necessary. Although peritoneal dialysis is viable for some, it is relatively contraindicated in those with significant ascites.

Kidney transplantation is a viable option in those who develop kidney failure, in the absence of major risk factors such as viral infection, undernutrition, or hepatic insufficiency. Uncomplicated residual hepatic fibrosis in the recipient does not seem to significantly modify the pharmacokinetics of immunosuppressive agents, but variations in cyclosporine blood levels can occur, perhaps secondary to altered GI absorption.⁵³ The severity of chronic liver disease may have a considerable impact on donor selection, choice of immunosuppression, and the eventual outcome.

Recurrence of schistosomal GN after transplantation has been described in a few patients,⁵³ suggesting the persistent release of antigens from living worms.⁵⁴ Although it is not an evidence-based practice, many authorities recommend the administration of a single dose of praziquantel before transplantation to recipients known to have been previously infected with the parasite.

SELF-ASSESSMENT QUESTIONS

1. A 45-year-old male recent Egyptian immigrant to Canada presents with a 4-week history of pyrexia, abdominal discomfort, asthenia, and increasing generalized edema. His medical history includes two treatment courses for intestinal schistosomiasis. Examination shows significant pallor, facial puffiness, a faint macular itchy eruption on the forearms and abdominal wall, and bilateral soft pedal edema. Temperature is 38.7°C (101.7°F), and blood pressure is 100/70 mm Hg. The liver edge is felt 2 cm below the right costal margin, firm and moderately tender. The spleen is felt 5 cm below the left costal margin and acutely tender. Urinary protein is 14 g/g creatinine, and the sediment shows 20 to 25 white blood cells, 8 to 10 red blood cells, and a few blood casts per high-power field. Peripheral blood hemoglobin is 8.6 g/dL; leukocyte count is 14,300/μL, of which 85% are neutrophils with left shift. Rheumatoid factor is positive at 48 IU/L, C-reactive protein is 128 units, and C3 is 54 mg/dL. Aspartate aminotransferase is 75 IU/mL, alanine aminotransferase is 62 IU/mL, and γ-glutamyl transferase is 123 IU/mL. Serum albumin is 1.8 g/dL, serum creatinine is 2.1 mg/dL, and blood urea nitrogen is 73 mg/dL. Which of the following is the *most* likely explanation for the patient's recent illness?
 - A. Ascending pyelonephritis complicating neglected urinary schistosomiasis
 - B. Schistosoma-associated *Salmonella* bacteremia
 - C. Active hepatitis viral infection on top of hepatointestinal schistosomiasis
 - D. Schistosoma-associated secondary immunoglobulin A nephropathy
 - E. Cryoglobulinemic vasculitis
2. A 50-year-old male Nigerian patient presents with severe burning during micturition, with passage of blood and necrotic tissue with urine for 2 weeks. He received treatment for urinary schistosomiasis

30 years previously and for recurrent urinary bacterial infection many times after that. Examination shows an ill man, with bilateral edema of the left lower limb. The left kidney is palpable and slightly tender. Rectal examination reveals a mass indenting the anterior rectal wall. Midstream urine examination shows a protein/creatinine ratio of 4 g/g, and uncountable red blood and pus cells. Serum creatinine is 4.3 mg/dL, and blood urea nitrogen is 103 mg/dL. The *most* likely cause of his edema is:

- A. Schistosomal glomerulonephritis
 - B. Schistosoma-associated chronic pyelonephritis
 - C. End-stage postobstructive kidney disease
 - D. Infiltrative bladder cancer
 - E. Advanced prostatic cancer
3. A 38-year-old female Brazilian patient presents with painful swelling of the small joints of her hands, right knee, and left shoulder; bilateral soft lower limb swelling; and a raised brownish-purple macular skin eruption over the tibial shins. She has been under medical follow-up for compensated chronic hepatosplenic schistosomiasis. Urine examination shows a protein-to-creatinine ratio of 5.2 g/g, and the sediment shows many red blood cells, few leukocytes, and red blood cell and granular casts. Serum creatinine is 2.1 mg/dL; blood urea nitrogen is 26 mg/dL; serum albumin is 2.7 g/dL; serum C3 is 86 mg/dL; and C4 is 3.2 mg/dL. Rheumatoid factor is 256 IU/mL. Alanine aminotransferase is 128 IU/L, aspartate aminotransferase is 9 IU/L, and γ-glutamyl transferase is 396 IU/L. Which of the following is the *most* likely diagnosis?
 - A. Acute schistosomiasis
 - B. Autoimmune hepatitis
 - C. Schistosomal glomerulonephritis
 - D. Concomitant virus-induced cryoglobulinemia
 - E. Superimposed rheumatoid arthritis

REFERENCES

- Ghalioungui P. *Magic and Medical Science in Ancient Egypt*. London: Hodder and Stoughton; 1963.
- Yang GJ, Vounatsou P, Zhou XN, et al. A review of geographic information system and remote sensing with applications to the epidemiology and control of schistosomiasis in China. *Acta Trop*. 2005;96:17–29.
- Abdel-Wahab MF. *Schistosomiasis in Egypt*. Boca Raton, FL: CRC Press; 1982.
- Barsoum RS. Schistosomiasis. In: Davison AM, Cameron JS, Grunfeld JP, et al., eds. *Oxford Textbook of Clinical Nephrology*. 3rd ed. Oxford: Oxford University Press; 2005:1173–1184.
- Kayange NM, Smart LR, Tallman JE, et al. Kidney disease among children in sub-Saharan Africa: systematic review. *Pediatr Res*. 2015;77(2):272–281.
- Neves PDMM, Jorge LB, Cavalcante LB, et al. Schistosomiasis-associated glomerulopathy: clinical aspects, pathological characteristics, and renal outcomes. *Clin Nephrol*. 2020;93(5):251–261.
- Dos-Santos WLC, Sweet GMM, Azevêdo LG, et al. Current distribution pattern of biopsy-proven glomerular disease in Salvador, Brazil, 40 years after an initial assessment. *J Bras Nefrol*. 2017;39(4):376–383.
- Chiamah OC, Ubachukwu PO, Anorue CO, et al. Urinary schistosomiasis in Ebonyi state, Nigeria from 2006 to 2017. *J Vector Borne Dis*. 2019;56(2):87–91.
- Wahl SM, Frazier-Jessen M, Jin WW, et al. Cytokine regulation of schistosome-induced granuloma and fibrosis. *Kidney Int*. 1997;51:1370–1375.
- Zheng B, Zhang J, Chen H, et al. T-Lymphocyte-Mediated liver Immunopathology of schistosomiasis. *Front Immunol*. 2020;11:61.
- Barsoum RS. Schistosomiasis and the kidney. *Semin Nephrol*. 2003;23:4–41.
- Hartgers FC, Smits HH, van der Kleij D, Yazdanbakhsh M. Innate, adaptive and regulatory responses in schistosomiasis: relationship to allergy. *Chem Immunol Allergy*. 2006;90:157–175.
- deWater R, Van Marck EA, Fransen JA, Deelder AM. Schistosoma mansoni: ultrastructural localization of the circulating anodic antigen and the circulating cathodic antigen in the mouse kidney glomerulus. *Am J Trop Med Hyg*. 1988;38:118–124.
- Abdel-Badie I, Barsoum RS. B-lymphocyte immunoglobulin surface expression in schistosomal glomerulopathy. *Med J Ahmed Maher Hosp*. 1999;1:1.
- Barsoum RS. Schistosomal glomerulopathies. *Kidney Int*. 1993;44:1–12.
- Houba V. Experimental renal disease due to schistosomiasis. *Kidney Int*. 1979;16:30–43.
- Gryseels B. The relevance of schistosomiasis for public health. *Trop Med Parasitol*. 1989;40:134–142.
- Mostafa M, Sheweita I, O'Connor P. Relationship between schistosomiasis and bladder cancer. *Clin Microbiol Rev*. 1999;12(1):97–111.
- Gouda I, Mokhtar N, Bilal D, et al. Bilharziasis and bladder cancer: a time trend analysis of 9843 patients. *J Egypt Natl Canc Inst*. 2007;19:158–162.
- Abd El-Aal A, Bayoumy I, Basyoni M, et al. Genomic instability in complicated and uncomplicated Egyptian schistosomiasis haematobium patients. *Mol Cytogenet*. 2015;8:1.
- Zaghloul MS, Zaghloul TM, Bishr MK, et al. Urinary schistosomiasis and the associated bladder cancer: update. *J Egypt Natl Canc Inst*. 2020;32:44.
- Nessim I, Francis M, Abdel-Rahman H, et al. Tubular proteins (TP) in hepatosplenic schistosomiasis (HSS). In: *Proceedings of the 2nd International Congress of Geographical Nephrology*. Egypt: Hurgada; 1993.
- Duarte DB, Vanderlei LA, RKdA B, et al. Renal function in hepatosplenic schistosomiasis – an assessment of renal tubular disorders. *PLoS One*. 2014;9(12):e115197.
- Duarte DB, Meneses GC, Lima DB, et al. Aquaporin-2 and NKCC2 expression pattern in patients with hepatosplenic. *Schistosomiasis*. 2020;25(9):1140–1144.
- Scheiner B, Lindner G, Reiberger T, et al. Acid-base disorders in liver disease. *J Hepatol*. 2017;67:1062–1073.
- Ezzat E, Osman R, Ahmed KY, Soothill JF. The association between Schistosoma haematobium infection and heavy proteinuria. *Trans R Soc Trop Med Hyg*. 1974;68:315–317.
- Barsoum R. The changing face of schistosomal glomerulopathy. *Kidney Int*. 2004;66:2472–2484.
- Barsoum R, Nabil M, Saady G, et al. Immunoglobulin A and the pathogenesis of schistosomal glomerulopathy. *Kidney Int*. 1996;50:920–928.
- Araújo SA, Neves PDMM, Wanderley DC, Reis MAD, et al. The immunohistological profile of membranous nephropathy associated with chronic Schistosoma mansoni infection reveals a glomerulopathy with primary features. *Kidney Int*. 2019;96(3):793–794.
- Neves PD, Bridi RA, Ramalho JA, et al. Schistosoma mansoni infection as a trigger to collapsing glomerulopathy in a patient with high-risk APOL1 genotype. *PLoS Negl Trop Dis*. 2020;14(10):e0008582.
- Hathout Sel -D, Abd el-Ghaffar Y, Awany AY. Salmonellosis complicating schistosomiasis in Egypt. A new clinical appreciation. *Am J Trop Med Hyg*. 1967;16(4):462–472.
- Kjetland EF1, Ndhlovu PD, Gomo E, et al. Association between genital schistosomiasis and HIV in rural Zimbabwean women. *AIDS*. 2006;20(4):593–600.
- Downs JA, van Dam GJ, Chagalucha JM, et al. Association of schistosomiasis and HIV infection in Tanzania. *Am J Trop Med Hyg*. 2012;87(5):868–873.
- Secor WE. The effects of schistosomiasis on HIV/AIDS infection, progression and transmission. *Curr Opin HIV AIDS*. 2012;7:254–259.
- Ganley-Leal LM, Mwinzi PN, Cetre-Sossah CB, et al. Correlation between eosinophils and protection against reinfection with Schistosoma mansoni and the effect of human immunodeficiency virus type 1 coinfection in humans. *Infect Immun*. 2006;74(4):2169–2176.
- Mbabazi PS, Andan O, Fitzgerald DW, et al. Examining the relationship between urogenital schistosomiasis and HIV infection. *PLoS Negl Trop Dis*. 2011;5(12):e1396.
- Al-Freih HM. Prevalence of hepatitis B surface antigenemia among patients with Schistosoma mansoni. *Ann Saudi Med*. 1993;13(2):121–125.
- Al-Shamiri AH, Al-Taj MA, Ahmed AS. Prevalence and co-infections of schistosomiasis/hepatitis B and C viruses among school children in an endemic areas in Taiz, Yemen. *Asian Pac J Trop Med*. 2011;4(5):404–408.
- Abruzzi A, Fried B, Alikhan SB. Coinfection of schistosoma species with hepatitis B or hepatitis C viruses. *Adv Parasitol*. 2016;91:111–231.
- Du H. Relationship between advanced schistosomiasis and HBV infection. *Zhongguo Xue Xi Chong Bing Fang Zhi Za Zhi*. 2013;25(1):115–117.
- Lemmer LB, Fripp PJ. Schistosomiasis and malignancy. *S Afr Med J*. 1994;84(3):211–215.
- Cooper K, Haffajee Z, Taylor L. Human papilloma virus and schistosomiasis associated bladder cancer. *Mol Pathol*. 1997;50:145–148.
- Swai B, Poggensee G, Mtweve S, et al. Female genital schistosomiasis as an evidence of a neglected cause for reproductive ill-health: a retrospective histopathological study from Tanzania. *BMC Infect Dis*. 2006;6:134.
- Weerakoon KG, Gobert GN, Cai P, McManus DP. Advances in the diagnosis of human schistosomiasis. *Clin Microbiol Rev*. 2015;28(4):939–967.
- Cnops L, Soentjens P, Clerinx J, Van Esbroeck M. A Schistosoma haematobium-specific real-time PCR for diagnosis of urogenital schistosomiasis in serum samples of international travelers and migrants. *PLoS Negl Trop Dis*. 2013;7:e2413.
- Bocanegra García C, Pintar Z, Serres X, Mendioroz J, et al. Ultrasound findings and associated factors to morbidity in Schistosoma haematobium infection in a highly endemic setting. *Trop Med Int Health*. 2018;23(2):221–228.
- Midzi N, Butterworth AE, Mdluzi T, et al. Use of circulating cathodic antigen strips for the diagnosis of urinary schistosomiasis. *Trans T Soc Trop Med Hyg*. 2009;103:45.
- Sandoval N, Siles-Lucas M, Pérez-Arellano JL, et al. A new PCR-based approach for the specific amplification of DNA from different Schistosoma species applicable to human urine samples. *Parasitology*. 2006;133:581.

49. Duarte DB, Lacerda MCSR, Ribeiro YJP, et al. Kidney biomarkers in tropical infections: an update. *Pathog Glob Health*. 2020;114(6):302–308.
50. Utzinger J, Tanner M, Keiser J. ACTs for schistosomiasis: do they act? *Lancet Infect Dis*. 2010;10(9):579–581.
51. Mahmoud KM, Sobh MA, El-Agroudy AE, et al. Impact of schistosomiasis on patient and graft outcome after renal transplantation: 10 years' follow-up. *Nephrol Dial Transplant*. 2001;16:2214–2221.
52. Valentim CL, Cioli D, Chevalier FD, et al. Genetic and molecular basis of drug resistance and species-specific drug action in schistosome parasites. *Science*. 2013;342(6164):1385–1389.
53. Azevedo LS, de Paula FJ, Ianhez LE, et al. Renal transplantation and schistosomiasis mansoni. *Transplantation*. 1987;44:795–798.
54. Kron M, Gordon C, Bauers T, et al. Persistence of schistosoma japonicum DNA in a kidney-liver transplant recipient. *Am J Trop Med Hyg*. 2019;100(3):584–587.